

TrES-1b

R-2331

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-1_120714 · created 2026-08-18T23:18:32+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2456122.94483 +/- 0.00094 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.002 +/- 0.052
Transit depth (Rp/Rs)^2	0.0004 +/- 0.021 [%]
Orbital Inclination (inc)	86.66 +/- 1.43
Airmass coefficient 1 (a1)	1.4 +/- 1.13
Airmass coefficient 2 (a2)	0.1 +/- 0.66
Scatter in the residuals of the lightcurve fit is	102.64 %
Best Comparison Star	#1 - [551, 426]
Optimal Aperture	0.75
Optimal Annulus	5.75
Transit Duration (day)	0.067 +/- 0.02

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.002 ± 0.052 vs 0.1358 (deviation 1.89σ)
Duration fit vs published	0.067 d vs 0.1042 d (deviation 1.37σ)
Mid-time O–C	-0.4 min
Depth SNR	0.0
Residual scatter	102.64 %

Light curve

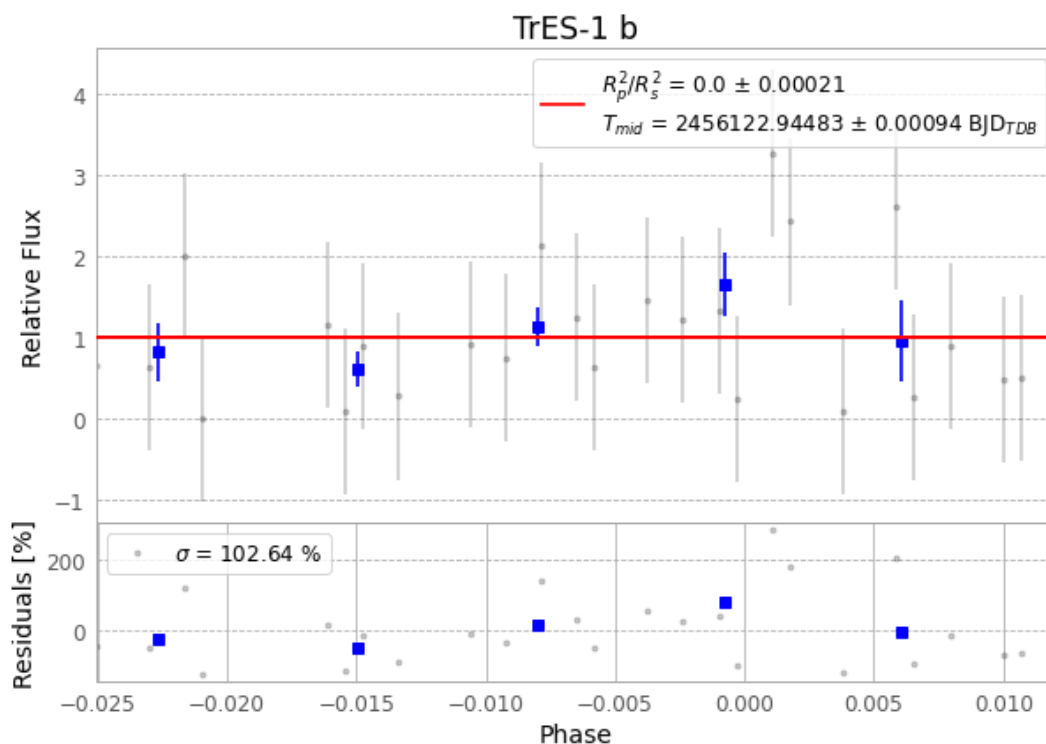


Figure 1 — FinalLightCurve_TrES-1 b_2012-07-14.png (SHA-256 46a065690c581901...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [277, 391], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 dfdd130f3398c99a...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

53 FITS frames (35 MB), TRES-1120714084513.FITS → TRES-1120714112713.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-1-120714.log (a1a65b57612b...), FinalLightCurve_TrES-1 b_2012-07-14.png (46a065690c58...), FinalLightCurve_TrES-1 b_2012-07-14.csv (e5dd6fde1019...)

Compute resources

Wall time 117.7 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-18 23:18Z.